

Order, Keys, and Yes/No: Auditing Option Bias in a Non-Autoregressive Decision Model

Sagar Maheshwari

Independent Researcher

maheshwari.sagars2000@gmail.com

Abstract

Non-autoregressive decision models answer typed questions, such as multi-class choices and yes/no questions, in a single encoder forward pass, and are designed to return calibrated probabilities for automated routing and gating. Option bias, meaning answers that depend on how options are presented rather than on the input, is well documented for generative language models but has not been studied systematically for this model class. We audit Laya, an open decision model, using its English (ModernBERT-based) and multilingual (mmBERT-based) checkpoints on four datasets and about 35,000 queries. We find that (i) on an easy four-way task both checkpoints are nearly invariant to option order, but on a harder six-way task consistency falls to 74.3% and 62.0%, and the multilingual checkpoint shows a significant recency bias (23.6% of choices go to the last slot versus 17.4% expected); (ii) when option keys conflict with their descriptions, the key wins on AG News in 80.5–92.0% of cases; (iii) yes/no framing is fragile: logically equivalent framings of one sentiment question contradict each other on 91–94% of SST-2 items, and under one wording the English checkpoint’s ranking is inverted (AUROC 0.352), which is invisible in the library’s default four-decimal output. In paired tests with multiple-comparison correction, zero-label contextual calibration raises accuracy by up to 25 points on fixed-template sentiment questions but never helps on BoolQ and significantly hurts there when probes are estimated per question, whereas Platt scaling on 200 labels never significantly reduced accuracy. We release code and raw outputs.

1 Introduction

A growing class of models answers structured questions without generating text. Given an input and a schema of typed questions, a *decision model* returns, for each question, a probability distribution

over its options in one encoder forward pass. Because such models are fast and output probabilities directly, they are positioned as components of automated pipelines: routing support tickets, flagging risk, or deciding whether an agent should act or escalate. In these settings the probabilities themselves drive actions, so they must depend on the input and not on incidental choices about how the question was written.

For generative large language models (LLMs), such incidental dependence is well documented. Multiple-choice answers shift with option order (Pezeshkpour and Hruschka, 2024), with the identity of option symbols such as “A” and “B” (Zheng et al., 2024), and with the order of in-context examples (Lu et al., 2022); few-shot predictions carry majority-label and recency biases that can be partly removed by calibration (Zhao et al., 2021). It is not obvious that these findings transfer to decision models. Their architecture differs in ways that plausibly matter: options are scored by a shared head at separate mask positions in a bidirectional encoder, and training uses strictly proper scoring rules intended to produce calibrated probabilities.

For the model we study, practitioners have reported individual failures on hand-built examples, including yes/no answers that follow the option labels rather than the input (Laya contributors, 2026b) and avoidance of the first-listed level on ordinal questions (ConvAI Innovations, 2026a). We present, to our knowledge, the first systematic, benchmark-scale audit of option bias in a non-autoregressive decision model. We study Laya (ConvAI Innovations, 2026b; Nandakishor M and Laya contributors, 2026), an open model with an English and a multilingual checkpoint, and probe three presentation factors: the *order* of options, the *keys* used to name them, and the *framing* of yes/no questions. For each we measure effects on both decisions and probabilities, and we test whether cheap inference-time fixes recover performance.

Our contributions are:

- A perturbation suite with item-level and paired statistics for auditing decision models, applied to two checkpoints and four datasets (AG News, DAIR Emotion, SST-2, BoolQ).
- Evidence that order robustness depends on task difficulty and checkpoint: near invariance on a four-way topic task, but weaker consistency on a six-way emotion task and a significant recency bias in the multilingual checkpoint.
- Evidence that option keys can override option descriptions, and that yes/no answers are highly sensitive to wording, including a hidden inversion of the English checkpoint’s ranking under one framing that is only visible at full output precision.
- A cautionary comparison of fixes: zero-label contextual calibration is effective only when the content-free probe measures pure bias, while light supervised calibration is consistently safe.

2 Background and Related Work

Decision models. Laya scores each option of each question at its own [MASK] position of a bidirectional encoder and applies a softmax over the option scores of a question (ConvAI Innovations, 2026b). The English checkpoint fine-tunes ModernBERT-large (Warner et al., 2024) and the multilingual checkpoint builds on mmBERT-base (Marone et al., 2025), each with a small decision head trained from scratch. Training uses reinforcement learning with a reward given by strictly proper scoring rules (Gneiting and Raftery, 2007), which in expectation reward reporting one’s true belief; it builds on the author’s earlier work on reinforcement learning for probability estimation (Nandakishor M, 2025b) and confidence-aware routing (Nandakishor M, 2025a). Questions are typed: *choice* (one of k named options, each with a key and a free-text description) and *noul* (a yes/no question returning $P(\text{true})$, whose two options are shown to the model with the labels `false` and `true`). Options and questions are defined at request time, so users choose the keys, descriptions and wording.

Option and order bias. LLM selection bias on multiple-choice questions has been attributed to priors over option-ID tokens and mitigated by estimating and removing that prior (Zheng et al., 2024); option-order sensitivity has been measured and linked to model uncertainty (Pezeshkpour and Hruschka, 2024). Few-shot prompting shows majority-label, recency and common-token biases (Zhao et al., 2021), and example order alone can move accuracy substantially (Lu et al., 2022). Surface-form competition (Holtzman et al., 2021) shows that the wording of answer strings affects scores independently of meaning. In cloze-style prompting of masked encoders, the choice of label words (verbalizers) strongly affects performance (Schick and Schütze, 2021; Gao et al., 2021); decision models inherit a related design choice through option keys and descriptions.

Calibration. We measure calibration with expected calibration error (ECE) (Naeini et al., 2015; Guo et al., 2017). Contextual calibration (Zhao et al., 2021) estimates a model’s bias from content-free inputs and divides it out, requiring no labels; later work refines the bias estimate (Fei et al., 2023; Zhou et al., 2024). Platt scaling (Platt, 1999) fits a logistic map on a small labelled set.

3 Experimental Setup

Model and environment. We use `laya` 0.3.22 with its default `english` and `multilingual` checkpoints on a single NVIDIA T4 GPU (transformers 5.0.0, PyTorch 2.10.0). The full run of about 35,000 queries took under 25 minutes. Newer library releases have since appeared (0.3.27 at the time of writing); they keep the default `noul` labels and add an opt-in override for them (Laya contributors, 2026a).

Output precision. The library rounds all returned probabilities to four decimal places. Because some of our findings concern probabilities below 10^{-4} , we disabled this rounding by overriding the module-level `round` used by the output formatter; this changes only reporting precision, not model computation. Section 6 shows why this matters.

Data. We use AG News (Zhang et al., 2015) (four topics; 200 test items), DAIR Emotion (Saravia et al., 2018) (six emotions; 300 test items), the SST-2 validation set (Socher et al., 2013; Wang et al., 2019) (872 items, 50.9% positive), and BoolQ

(Clark et al., 2019) (800 validation items; 59.9% “yes”). Samples are drawn with a fixed seed. Option descriptions and instructions (Appendix A) are our own and are held fixed within each experiment, so only the manipulated factor changes.

Perturbations. *Order:* each AG News item is asked under all 24 option orderings and each Emotion item under 6 random orderings. *Keys:* options are renamed with semantic keys (e.g. *sports*), letters, numbers, random strings, or *misleading* keys, where each description is paired with the name of a different class. *Yes/no framing:* each SST-2 review is asked as `noul` “Is this review positive?”, as `noul` “Is this review negative?” (scored as $1 - p$), and as a two-option choice with neutral keys A and B and yes/no descriptions, with “yes” as option A or as option B. The two-option form is the workaround recommended in the model card when `noul` answers look unreliable (ConvAI Innovations, 2026b). BoolQ questions are asked as `noul` and as the two-option choice in both orders.

Metrics and statistics. We report accuracy, top-label ECE with 15 equal-width bins, and AUROC for binary tasks. For order we report *permutation consistency* (fraction of items answered identically under all orderings), the share of choices per slot compared with the share of correct answers per slot, and the fraction of items whose probability on the correct option $p(\text{gold})$ varies by more than 0.2 across orderings. Confidence intervals for individual metrics are 95% percentile bootstraps with 1,000 resamples (Efron and Tibshirani, 1993); for order, whole items are resampled because orderings of the same item are dependent. Comparisons between conditions evaluated on the same items use an exact two-sided McNemar test on per-item correctness (McNemar, 1947), with Holm–Bonferroni correction (Holm, 1979) within each family of comparisons (38 calibration comparisons; 12 key-scheme comparisons), and paired bootstrap intervals (2,000 resamples) for differences in accuracy and ECE. We call a difference significant when its Holm-adjusted $p < 0.05$.

4 Order

Table 1 and Figure 1 summarise the order experiments. On AG News both checkpoints are close to order-invariant: 95.5% (English) and 94.0% (multilingual) of items receive the same answer under all 24 orderings, each slot is chosen almost ex-

actly as often as it holds the correct answer, and averaging predictions over orderings does not improve accuracy. The English checkpoint shows a statistically detectable but negligible primacy effect (slot 1 chosen 25.4%, interval [25.1, 25.8]; slot 4 chosen 24.6%, [24.2, 24.9]). This contrasts with the strong option-ID priors reported for generative LLMs (Zheng et al., 2024), consistent with the shared per-option scorer making the architecture largely order-agnostic.

Order sensitivity grows with task difficulty. On the six-way Emotion task, consistency falls to 74.3% (English) and 62.0% (multilingual), and $p(\text{gold})$ swings by more than 0.2 for 28.7% and 45.3% of items. For the English checkpoint this instability is undirected: every slot’s interval contains its expected share. The multilingual checkpoint instead shows a clear *recency bias*: 23.6% of its choices go to the last slot (interval [21.5, 25.9]) against 17.4% expected, and slots 1 and 2 are under-selected (14.1% and 14.4%). This is consistent with a previously reported avoidance of the first-listed level on ordinal questions for the same checkpoint (ConvAI Innovations, 2026a); our results show that a positional preference also affects choice questions. Accuracy is correspondingly higher when the correct answer is listed last (0.591) than when it is listed second (0.399). Averaging over orderings did not recover accuracy (0.491 vs. 0.487), suggesting the order effect acts as noise on uncertain items rather than a bias that averaging can cancel.

5 Option Keys

Table 2 shows that when keys carry no conflicting meaning, their form matters little for accuracy. Letters, numbers and random strings agree with semantic-key answers on 97.0–97.5% of AG News items, and paired tests find no significant accuracy difference from semantic keys in any of the 12 comparisons after correction. The largest uncorrected effect is on multilingual Emotion, where letters and numbers outperform semantic keys by 6.3 and 6.0 points (McNemar $p = 0.005$ and 0.015 ; Holm-adjusted 0.065 and 0.169). The model can therefore work from descriptions alone.

When keys *conflict* with descriptions, however, keys often dominate. On AG News, 80.5% (English) and 92.0% (multilingual) of answers follow the key name, and accuracy collapses to 0.165 and 0.050. On Emotion the effect is weaker but still

Checkpoint	Dataset	Orderings	Consistency	Swing > 0.2	Acc. (single)	Acc. (averaged)	Max slot deviation	$\chi^2 p$
English	AG News (4)	24	0.955	0.110	0.923	0.920	0.004	0.875
English	Emotion (6)	6	0.743	0.287	0.579	0.593	0.018	0.162
Multilingual	AG News (4)	24	0.940	0.095	0.946	0.945	0.001	0.996
Multilingual	Emotion (6)	6	0.620	0.453	0.491	0.487	0.062	9×10^{-11}

Table 1: Order sensitivity. *Consistency*: items answered identically under all orderings. *Swing*: items whose $p(\text{gold})$ varies by more than 0.2 across orderings. *Max slot deviation*: largest gap between a slot’s share of choices and its share of correct answers. The χ^2 test treats calls as independent and therefore overstates significance; the item-bootstrap intervals in Figure 1 are the primary evidence.

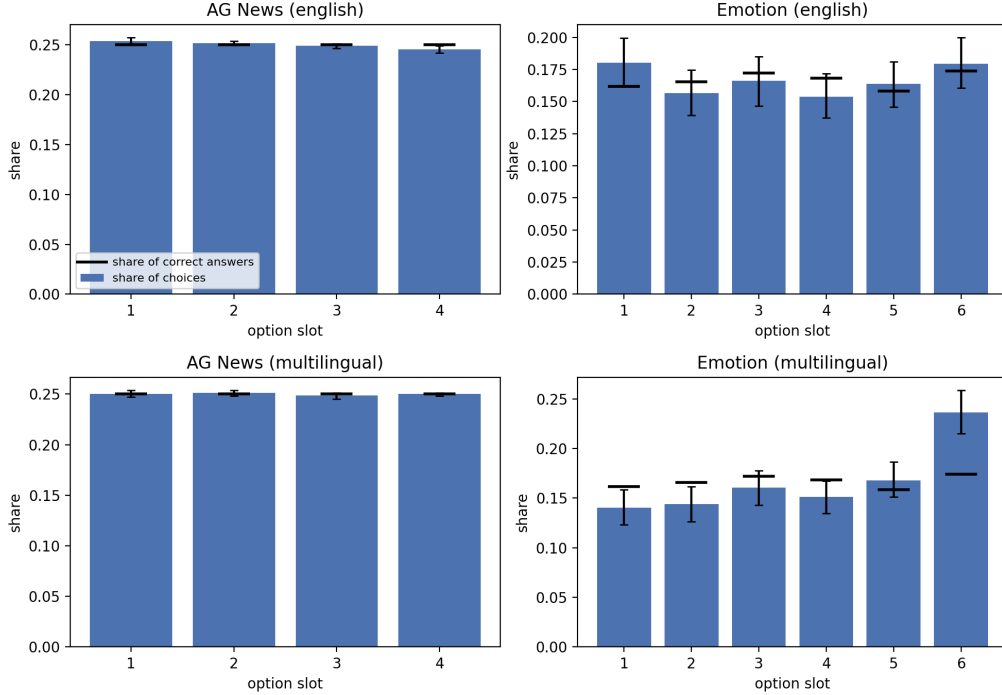


Figure 1: Share of choices per option slot (bars; 95% item-bootstrap intervals) against the share of correct answers per slot (black ticks). Only the multilingual checkpoint on Emotion deviates clearly, over-selecting the last slot.

damaging: accuracy falls from 0.597 to 0.393 (English) and from 0.483 to 0.167 (multilingual), with 24.3% and 37.3% of answers following the key. While misleading keys are constructed, the situation is realistic under *schema drift*, when a description is updated but a legacy key is kept, or when internal codes happen to be meaningful words.

Keys also affect probabilities without changing decisions. With random-string keys, the English checkpoint on AG News keeps its accuracy (0.925) but its mean $p(\text{gold})$ falls from 0.839 to 0.726 and ECE rises from 0.063 to 0.172; this effect does not appear for the multilingual checkpoint or on Emotion. Conversely, on multilingual Emotion, replacing semantic keys with letters or numbers roughly halves ECE (0.304 to 0.143 and 0.126).

6 Yes/No Framing

Table 3 compares four logically equivalent ways of asking whether an SST-2 review is positive. The `noul` framings fail as classifiers, quantifying at benchmark scale a failure first reported on hand-built examples (Laya contributors, 2026b). Asked “Is this review positive?”, both checkpoints answer “no” for nearly every review (recall on positive reviews 0.000 and 0.009); asked “Is this review negative?”, they also mostly answer “no”. As a result, the two `noul` framings contradict each other on 90.8% (English) and 94.4% (multilingual) of reviews, and on 76.7% and 86.9% of reviews the model states with $P < 0.2$ that the review is *neither* positive nor negative. Content-free probes confirm a strong prior toward “no” in both `noul` framings.

In all but one case, the failure is a shifted thresh-

Key scheme	AG News, English		AG News, multilingual		Emotion, English		Emotion, multilingual	
	Acc.	ECE	Acc.	ECE	Acc.	ECE	Acc.	ECE
Semantic	0.920	0.063	0.950	0.033	0.597	0.218	0.483	0.304
Letters	0.925	0.048	0.940	0.032	0.580	0.236	0.547	0.143
Numbers	0.920	0.048	0.935	0.037	0.590	0.226	0.543	0.126
Random strings	0.925	0.172	0.945	0.035	0.610	0.187	0.493	0.149
Misleading	0.165	0.538	0.050	0.900	0.393	0.165	0.167	0.551
<i>Misleading keys: share of choices following the key name rather than the description</i>								
	0.805		0.920		0.243		0.373	

Table 2: Effect of option keys with fixed descriptions and fixed order. Under *misleading* keys each description carries another class’s name. Accuracy intervals are ± 0.03 – 0.06 (Appendix B).

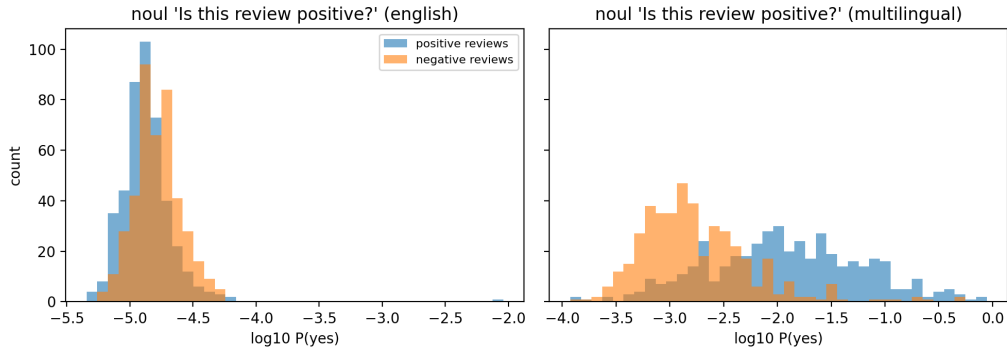


Figure 2: $\log_{10} P(\text{yes})$ for the `noul` question “Is this review positive?” on SST-2, split by true label. For the English checkpoint (left) all values lie between roughly $10^{-5.3}$ and $10^{-4.2}$ (one outlier near 10^{-2}), below the library’s display precision, and negative reviews score slightly *higher* than positive ones. The multilingual checkpoint (right) separates the classes in the correct direction.

Ckpt	Framing	Acc.	AUROC	CF $P(\text{pos})$
En	<code>noul</code> “positive?”	0.491	0.352	0.000
	<code>noul</code> “negative?”	0.599	0.933	1.000
	choice, yes = A	0.795	0.944	0.098
	choice, yes = B	0.750	0.937	0.034
	choice, averaged	0.773	0.948	0.066
MI	<code>noul</code> “positive?”	0.494	0.830	0.001
	<code>noul</code> “negative?”	0.553	0.896	0.998
	choice, yes = A	0.716	0.910	0.004
	choice, yes = B	0.819	0.919	0.091
	choice, averaged	0.776	0.919	0.048

Table 3: SST-2 ($n=872$) under equivalent yes/no framings, all scored as $P(\text{positive})$. *CF*: output on content-free inputs; an unbiased framing would give 0.5.

old rather than a failure to read the input: AUROC stays between 0.830 and 0.948, meaning reviews are ranked correctly but placed on the wrong side of 0.5. The exception is the English checkpoint’s “positive?” framing, whose AUROC is 0.352 (interval $[0.318, 0.390]$): its ranking is *inverted*. All its probabilities lie between 4.5×10^{-6} and 9.1×10^{-3} (median 1.4×10^{-5} ; Figure 2), so under the library’s default four-decimal rounding most outputs

display as exactly 0, and in a preliminary run on rounded outputs this framing appeared to carry no signal (AUROC 0.50). The inversion only becomes visible at full precision. The multilingual checkpoint does not share it (AUROC 0.830).

The recommended two-option choice framing is more usable (accuracy 0.716–0.819) but does not remove the bias: it still leans toward “no” (recall on negative reviews 0.907–0.984 versus 0.471–0.734 on positive ones) and depends on option order by 4.5 points (English) to 10.3 points (multilingual). The direction of the order effect is not consistent: the English checkpoint does better with “yes” first, the multilingual checkpoint with “yes” second. On BoolQ, by contrast, the `noul` framing works well for the English checkpoint (test accuracy 0.780 against a 0.595 majority baseline, ECE 0.044; Table 4), so the SST-2 failure depends on question wording rather than being a general property of `noul` questions.

7 Mitigation

We compare two inference-time fixes on held-out test splits (Table 4; Figure 3 in the appendix). *Con-*

Task	Ckpt	Framing	Raw	Contextual (global)	Contextual (per item)	Platt (200 labels)
SST-2	En	noul “positive?”	0.488 (0.512)	0.448 (0.175)	–	0.625 (0.068)
		noul “negative?”	0.603 (0.307)	0.762 (0.179)	–	0.835 (0.057)
		choice, yes = A	0.777 (0.113)	0.869 (0.028)	–	0.868 (0.031)
		choice, yes = B	0.735 (0.139)	0.832 (0.044)	–	0.862 (0.025)
		choice, averaged	0.754 (0.116)	0.874 (0.050)	–	0.876 (0.026)
	Ml	noul “positive?”	0.494 (0.484)	0.592 (0.216)	–	0.756 (0.037)
		noul “negative?”	0.552 (0.409)	0.802 (0.063)	–	0.795 (0.052)
		choice, yes = A	0.707 (0.198)	0.677 (0.172)	–	0.833 (0.032)
		choice, yes = B	0.814 (0.062)	0.823 (0.070)	–	0.836 (0.030)
		choice, averaged	0.772 (0.088)	0.811 (0.069)	–	0.836 (0.048)
BoolQ	En	noul	0.780 (0.044)	0.728 (0.126)	0.675 (0.189)	0.780 (0.051)
		choice, yes = A	0.768 (0.061)	0.770 (0.066)	0.727 (0.102)	0.790 (0.030)
		choice, yes = B	0.783 (0.066)	0.743 (0.115)	0.693 (0.150)	0.803 (0.039)
	Ml	noul	0.655 (0.206)	0.643 (0.269)	0.610 (0.311)	0.658 (0.040)
		choice, yes = A	0.637 (0.285)	0.627 (0.295)	0.620 (0.295)	0.618 (0.040)
		choice, yes = B	0.660 (0.091)	0.642 (0.201)	0.627 (0.217)	0.655 (0.032)

Table 4: Test accuracy (ECE in parentheses) after inference-time calibration. Test sets: SST-2 $n=672$, BoolQ $n=600$ (majority baseline 0.595); Platt scaling is fit on a disjoint set of 200 labelled items. Bold: significant improvement over raw; red: significant degradation (exact McNemar test on the same items, Holm-corrected over all 38 comparisons, adjusted $p < 0.05$). SST-2 contextual calibration uses one prior from three content-free inputs; for BoolQ we compare one prior averaged over the calibration split (global) with each question’s own content-free probe (per item).

textual calibration (Zhao et al., 2021) queries the model with the question and a content-free input (“N/A”, “[MASK]”, “...”), treats the result as the model’s prior, and subtracts it in logit space; it needs no labels. *Platt scaling* fits a logistic regression on the logit of the model’s probability using 200 labelled items.

Contextual calibration helps on SST-2 but not on BoolQ. On SST-2 it significantly improves accuracy in 5 of 10 settings with no significant degradation; for the English choice framing, accuracy rises from 0.777 to 0.869 (paired difference +0.092, interval [+0.062, +0.122]) and ECE falls from 0.113 to 0.028, matching Platt scaling without using labels. For multilingual noul “negative?”, accuracy rises by 25 points (0.552 to 0.802). It cannot repair the English inverted framing, since a monotone shift preserves the reversed ranking. On BoolQ it never significantly improves accuracy. With a single global prior, no change survives correction (the largest, English noul 0.780 to 0.728, has uncorrected $p = 0.003$, adjusted 0.056), and it significantly worsens ECE in 5 of 6 settings (paired bootstrap). With each question’s own probe it significantly lowers accuracy for all three English framings (e.g. noul 0.780 to 0.675).

Two mechanisms plausibly explain the BoolQ result. First, English content-free probes carry some knowledge of the answer: their AUROC against the

gold label is 0.556–0.578, with intervals excluding 0.5, so subtracting them removes signal. Second, per-question probes are noisy estimates of bias: multilingual probes carry no detectable knowledge (AUROC 0.505–0.529), yet per-item correction still lowers accuracy in all three multilingual settings, and per-item correction is worse than a global prior in all six BoolQ settings. More broadly, on SST-2 the sentiment question is meaningless without a review, so the content-free answer measures pure bias; on BoolQ, answering “no” when the passage is “N/A” is arguably reasonable behaviour, and the raw English outputs are already well calibrated (ECE 0.044–0.066), leaving little bias to remove.

Light supervision is consistently safe. Platt scaling improved accuracy significantly in 9 of 10 SST-2 settings and never significantly reduced accuracy anywhere; on BoolQ, changes ranged from -0.018 to $+0.022$, and the one uncorrected decrease (multilingual choice, $p = 0.035$) does not survive correction. It reduced ECE in 15 of 16 settings, significantly in 13 (paired bootstrap), the exception being English BoolQ noul, which was already well calibrated (0.044 to 0.051). Because it can learn a negative slope, Platt scaling also exploits the inverted English framing, raising its accuracy from 0.488 to 0.625 (fitted slope -0.705). It cannot raise AUROC, however: on BoolQ the multilingual checkpoint’s discrimination is limited (AUROC

0.641–0.706) and no calibration method lifts accuracy meaningfully above the majority baseline.

8 Discussion

Where bias lives. In this architecture, bias lives mainly in the *semantics* of the question and option text rather than in *position*. Order effects are negligible on an easy task and grow as decision margins shrink, and only one checkpoint shows a directional preference. Key names and yes/no wording, in contrast, can override the content of descriptions and inputs. We hypothesise that a shared scorer applied at each option’s mask position makes position largely irrelevant, while the text of keys and questions directly shapes the representation at that position; testing this mechanistically is left to future work.

Calibration can hide failures. Aggregate calibration metrics can look good while the model is wrong in structured ways. After contextual calibration, the English inverted framing reaches ECE 0.175 at 0.448 accuracy, and a model that outputs about 0.5 for everything would be nearly perfectly calibrated while useless. We therefore recommend always reporting discrimination (AUROC) and accuracy alongside ECE. Likewise, output rounding is not harmless: a four-decimal display turned an inverted ranking into apparent noise. Measurements of label effects that inform library defaults should therefore use raw probabilities or logits, as the maintainers have also requested (Laya contributors, 2026a).

Recommendations for practitioners. Based on these results, we suggest: (1) avoid option keys that are meaningful words unless they match the descriptions, and prefer neutral keys when descriptions carry the meaning; (2) for binary decisions, prefer a two-option choice over `noul`, but still validate the wording and option order on labelled data; (3) when a small labelled set is available, fit Platt scaling or temperature scaling, which was never harmful here; (4) use contextual calibration only for fixed-template questions whose content-free answer should be uninformative; (5) for tasks with many similar options, test order robustness on the target checkpoint.

Future work. Each yes/no framing was tested with a single wording; testing several paraphrases of each question would establish how general the inverted ranking is. Training-time interventions, such

as randomising option order, keys and `noul` label words during fine-tuning, may remove these biases at their source. Extending the audit to ordinal *score* questions, to the opt-in `noul` label override (Laya contributors, 2026a), and to other decision models would test how far our findings generalise.

Limitations

We study one model family and two checkpoints, so our conclusions may not transfer to other decision models. Our results use `laya` 0.3.22 and the checkpoints it loaded by default; later releases or re-trained checkpoints may behave differently. Option descriptions and question wording were written by us, and each yes/no framing was tested with one wording; different wording could change absolute results, and the SST-2 findings show that wording effects can be large. The model card reports results on some of the datasets we use, so the checkpoints may have been trained or tuned on their training splits; we evaluate only on test and validation splits, but cannot rule out effects of familiarity with these distributions. Sample sizes (200–872 items per condition) give intervals of several points; paired tests with multiple-comparison correction are applied to the calibration and key-scheme comparisons, but other reported intervals are uncorrected. The multi-lingual checkpoint was evaluated only on English data. Misleading keys are an adversarial construction, although we argue they model schema drift. Our content-free probes are one of several possible choices (Fei et al., 2023; Zhou et al., 2024). We evaluate only inference-time fixes. Finally, we disabled the library’s output rounding for measurement, which deployed users would not normally do.

Ethics Statement

This work audits a publicly available model on public datasets and involves no human subjects. Our findings concern reliability of automated decisions; we hope they help practitioners avoid over-trusting model probabilities in consequential settings.

Use of AI Assistants

The author used Claude (Anthropic), an AI assistant, extensively in this project, including for literature search, experiment design, writing the experiment and analysis code, statistical analysis, and drafting and editing the manuscript. The author ran all experiments, reviewed the code, analyses and

text, and takes full responsibility for the content of this paper.

References

- Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. 2019. BoolQ: Exploring the surprising difficulty of natural yes/no questions. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*.
- ConvAI Innovations. 2026a. Laya multilingual (hugging face model card). <https://huggingface.co/convaiinnovations/laya-multilingual>. Accessed October 2026.
- ConvAI Innovations. 2026b. Laya: Non-autoregressive decision model (hugging face model card). <https://huggingface.co/convaiinnovations/laya>. Accessed October 2026.
- Bradley Efron and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*. Chapman and Hall.
- Yu Fei, Yifan Hou, Zeming Chen, and Antoine Bosselut. 2023. Mitigating label biases for in-context learning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*.
- Tianyu Gao, Adam Fisch, and Danqi Chen. 2021. Making pre-trained language models better few-shot learners. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*.
- Tilman Gneiting and Adrian E. Raftery. 2007. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*.
- Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.
- Ari Holtzman, Peter West, Vered Shwartz, Yejin Choi, and Luke Zettlemoyer. 2021. Surface form competition: Why the highest probability answer isn’t always right. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*.
- Laya contributors. 2026a. fix(agent): allow custom noul labels. GitHub pull request #163, <https://github.com/NandhaKishorM/laya/pull/163>.
- Laya contributors. 2026b. noul returns the negative label regardless of state: a true/false or yes/no label pair decides the answer. GitHub issue #156, <https://github.com/NandhaKishorM/laya/issues/156>.
- Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenetorp. 2022. Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*.
- Marc Marone, Orion Weller, William Fleshman, Eugene Yang, Dawn Lawrie, and Benjamin Van Durme. 2025. mmBERT: A modern multilingual encoder with annealed language learning. *arXiv preprint arXiv:2509.06888*.
- Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.
- Mahdi Pakdaman Naeini, Gregory F. Cooper, and Milos Hauskrecht. 2015. Obtaining well calibrated probabilities using Bayesian binning. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Nandakishor M. 2025a. Confidence-aware routing for large language model reliability enhancement: A multi-signal approach to pre-generation hallucination mitigation. *arXiv preprint arXiv:2510.01237*.
- Nandakishor M. 2025b. SalesRLAgent: A reinforcement learning approach for real-time sales conversion prediction and optimization. *arXiv preprint arXiv:2503.23303*.
- Nandakishor M and Laya contributors. 2026. Laya: Fast, non-autoregressive system 1 decision engine. <https://github.com/NandhaKishorM/laya>.
- Pouya Pezeshkpour and Estevam Hruschka. 2024. Large language models sensitivity to the order of options in multiple-choice questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*.
- John C. Platt. 1999. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*. MIT Press.
- Elvis Saravia, Hsien-Chi Toby Liu, Yen-Hao Huang, Junlin Wu, and Yi-Shin Chen. 2018. CARER: Contextualized affect representations for emotion recognition. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*.
- Timo Schick and Hinrich Schütze. 2021. Exploiting cloze-questions for few-shot text classification and natural language inference. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*.
- Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Ng, and Christopher Potts. 2013. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*.

Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. 2019. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In *International Conference on Learning Representations*.

Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Nathan Cooper, Griffin Adams, Jeremy Howard, and Iacopo Poli. 2024. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. *arXiv preprint arXiv:2412.13663*.

Xiang Zhang, Junbo Zhao, and Yann LeCun. 2015. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems*.

Zihao Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. 2021. Calibrate before use: Improving few-shot performance of language models. In *Proceedings of the 38th International Conference on Machine Learning*.

Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. 2024. Large language models are not robust multiple choice selectors. In *The Twelfth International Conference on Learning Representations*.

Han Zhou, Xingchen Wan, Lev Prolee, Diana Mincu, Jilin Chen, Katherine Heller, and Subhrajit Roy. 2024. Batch calibration: Rethinking calibration for in-context learning and prompt engineering. In *The Twelfth International Conference on Learning Representations*.

A Prompts and Option Descriptions

AG News. Instruction: “What is the topic of this news article?” Descriptions: `world`: “world news, politics, international affairs”; `sports`: “sports, games, athletes, teams”; `business`: “business, companies, markets, economy”; `scitech`: “science and technology, computers, research”.

Emotion. Instruction: “Which emotion does the writer express?” Descriptions: `sadness`: “sad, unhappy, grieving, lonely”; `joy`: “happy, cheerful, pleased, excited”; `love`: “love, affection, caring, romantic feelings”; `anger`: “angry, annoyed, irritated, furious”; `fear`: “afraid, scared, anxious, nervous”; `surprise`: “surprised, amazed, astonished, shocked”.

Key schemes. Letters A, B, C, ...; numbers 1, 2, 3, ...; random strings `qzv`, `lmt`, `wkr`, `pfy`, `dxo`, `bnu`; misleading keys assign class i ’s description the name of class $i+1$ (cyclically).

SST-2. `noul` instructions “Is this review positive?” and “Is this review negative?”, using the library’s default `noul` labels. Choice instruction “Is this review positive?” with options “yes, the review is positive” and “no, the review is negative” under keys A and B in both orders.

BoolQ. The passage is the input and the question (capitalised, with a question mark) is the instruction. Choice options are “yes” and “no” under keys A and B in both orders. Content-free probes replace the passage with “N/A”.

B Reproducibility

All experiments are produced by a single script with a fixed seed (0), which caches every raw model output as JSONL and prints all tables with 95% bootstrap intervals; a second script computes the paired tests from the cached outputs without further model calls. Accuracy intervals for Table 2 range from ± 0.03 (AG News) to ± 0.06 (Emotion). The released outputs contain item indices, labels and model probabilities but no dataset text. Code and raw outputs: <https://github.com/maheshwari2000/laya-bias-audit>.

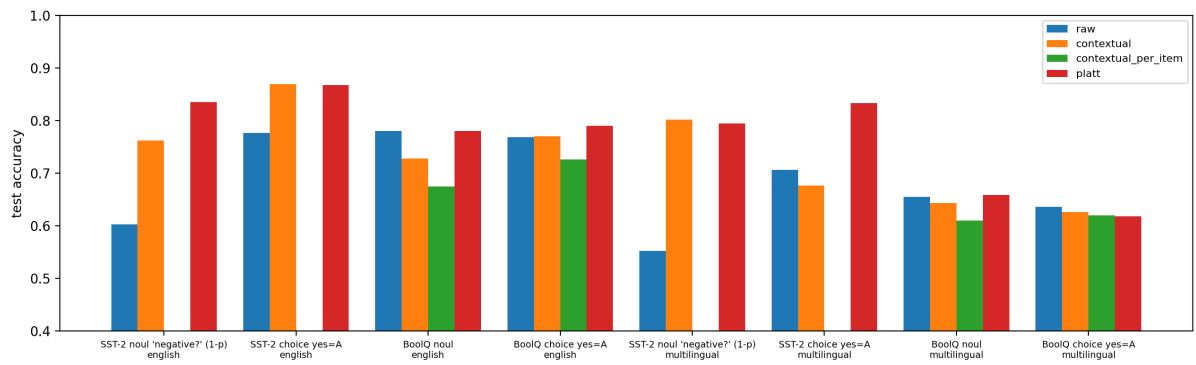


Figure 3: Test accuracy by calibration method for selected settings (numbers in Table 4). Missing green bars: per-item contextual calibration applies only to BoolQ.